

A Real-time Research Project Report

on

AGROSENSE-Price Prediction System

submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE & ENGINEERING

by

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BVRIT HYDERABAD College of Engineering for Women

(NAAC Accredited-A Grade | NBA Accredited B.Tech. (EEE, ECE & CSE))

(UGC Autonomous, Approved by AICTE, New Delhi and Affiliated to JNTUH, Hyderabad)

Bachupally, Hyderabad – 500090

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CERTIFICATE

This is to certify that the Real-time Research Project report on “**AGROSENSE**” is a bonafide work carried out by **Ms. M.Lakshmi Tanooja(24WH1A05E1), Ms.M.Greeshmitha (24WH1A05E2) and Ms. M.Godha Sree (24WH1A05J5)** in the partial fulfillment for the award of B.Tech. degree in **Computer Science & Engineering, BVRIT HYDERABAD College of Engineering for Women**, Bachupally, Hyderabad, affiliated to Jawaharlal Nehru Technological University Hyderabad, Hyderabad under my guidance and supervision. The results embodied in the real-time research project work have not been submitted to any other University or Institute for the award of any degree or diploma.

Internal Guide

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DECLARATION

We hereby declare that the work presented in this Real-time Research Project entitled **“AGROSENSE - Price Prediction System”** submitted towards completion of Real-time research project work in II B.Tech., CSE at ‘BVRIT HYDERABAD College of Engineering for Women’, Hyderabad is an authentic record of our original work carried out under the guidance of **Ms. KBKS Durga**, Assistant Professor, Department of CSE.

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ABSTRACT

Agricultural commodity prices are highly volatile due to seasonal variations, climatic conditions, demand supply changes, and market fluctuations. Farmers often sell their produce without prior knowledge of future price trends, which can lead to financial losses. Existing agricultural platforms mainly provide historical price information but lack predictive analysis to support informed decision-making.

This project, AgroSense, presents a crop price prediction system using time-series forecasting techniques. The system collects historical agricultural market price data from publicly available sources using web scraping methods. The collected data is stored and preprocessed using Python libraries such as Pandas and NumPy to remove inconsistencies and prepare the dataset for analysis.

The AutoRegressive Integrated Moving Average (ARIMA) model is applied to analyze historical price patterns and forecast future crop prices. The predicted results are generated through a Python based implementation and exposed via a REST API for accessing forecast data. Automation tools are used to manage workflow updates and ensure continuous data processing.

By integrating automated data collection, statistical forecasting, and machine learning techniques, AgroSense aims to provide reliable crop price predictions that help farmers and stakeholders make better data-driven decisions and reduce uncertainty in agricultural markets. findings or lessons learned.

Keywords: *Agricultural Price Prediction, ARIMA Model, Time Series Forecasting, Machine Learning, Crop Price Analysis, Web Scraping, Data Preprocessing, Agricultural Market Analysis, Predictive Analytics.*

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LIST OF TERMS AND ABBREVIATIONS

ACRONYM	EXPANSION
AL	Artificial Intelligence
ML	Machine Learning
ARIMA	AutoRegressive Integrated Moving Average
TS	Time Series
API	Application Programming Interface
DF	Data Frame
CSV	Comma Separated Values
CSE	Computer Science and Engineering
BVRITH	BVRIT Hyderabad College of Engineering for Women
JNTUH	Jawaharlal Nehru Technological University Hyderabad

CHAPTER 1

INTRODUCTION

1.1 Background

Agriculture is one of the most fundamental sectors of the Indian economy, playing a crucial role in sustaining livelihoods and supporting national development. A significant portion of the population depends directly or indirectly on agriculture for income and employment. Beyond its economic contribution, agriculture is essential for ensuring food security for a rapidly growing population. It provides raw materials to various industries such as textiles, food processing, and pharmaceuticals, thereby strengthening the overall industrial ecosystem and contributing to the country's GDP. In addition to its economic importance, agriculture holds strong social and cultural significance in India. It forms the backbone of rural communities and plays a key role in shaping traditions, lifestyles, and regional practices. The sector supports a wide range of activities including crop cultivation, livestock management, and allied industries, creating a diverse and interconnected system. With advancements in irrigation, fertilizers, and farming techniques, agriculture has evolved over time, improving productivity and contributing to the nation's growth and self-sufficiency. In recent years, the integration of technology into agriculture has opened new opportunities for modernization and efficiency. Concepts such as precision farming, data-driven decision-making, and smart agriculture are transforming traditional practices into more advanced and sustainable systems. The use of technologies like data analytics and machine learning is enabling better resource management, improved productivity, and enhanced planning. This shift towards technology-driven agriculture highlights the growing importance of innovation in strengthening the sector and ensuring its long-term sustainability.

1.2 Problem Statement

Farmers often face uncertainty in deciding the right time to sell their crops due to the absence of accurate and reliable price prediction systems. Crop prices vary significantly over time because of changing market conditions, seasonal patterns, and demand–supply dynamics. In the current scenario, farmers mainly rely on traditional knowledge, past experiences, or limited available information to make selling decisions, which may not always reflect future market trends. Most existing systems provide only historical or current price data and lack the capability to predict future price movements. Without access to predictive insights, farmers are unable to analyze upcoming market conditions effectively. This results in decision-making based on incomplete information, reducing their ability to plan selling strategies and take advantage of favorable price periods. Therefore, the core problem lies in the lack of an intelligent and data-driven system that can analyze historical trends and accurately forecast future crop prices. Addressing this gap is essential to support better decision-making, reduce uncertainty, and improve overall efficiency in agricultural market participation.

1.3 Proposed System

The proposed system, AgroSense, is designed to provide an intelligent solution for predicting future crop prices using data-driven techniques. The system focuses on analyzing historical agricultural price data and identifying patterns and trends over time. By transforming raw data into a structured format and applying preprocessing techniques, the system ensures that the data is clean, consistent, and suitable for accurate analysis. To generate reliable predictions, the system utilizes time-series forecasting methods, specifically the ARIMA model, which is effective in capturing patterns such as trends and fluctuations in sequential data. The model is trained on historical price data to understand past behavior and then used to forecast future price values. This enables the system to provide short-term predictions that can help users anticipate market trends more effectively. The predicted results are presented through clear and interactive visualizations, allowing users to compare historical data with forecasted values. By providing accurate and easy-to-understand insights, the system supports informed decision-making and helps in planning better selling strategies. Overall, AgroSense aims to bridge the gap between raw agricultural data and meaningful predictions by combining data processing, forecasting, and visualization into a single integrated system.

1.4 Objectives

The primary objective of the AgroSense project is to design and develop an efficient machine learning-based system that can accurately predict future crop prices using historical market data. The system aims to utilize time-series forecasting techniques to analyze past price trends and generate reliable predictions that reflect possible future market behavior. By focusing on accuracy and consistency, the project intends to provide meaningful insights that can be trusted for decision making purposes. Another key objective is to automate the entire data processing pipeline, including data collection, preprocessing, and analysis. The system is designed to handle raw data by cleaning inconsistencies, organizing it into a structured format, and transforming it into a suitable form for forecasting. This automation reduces manual effort, improves efficiency, and ensures that the data used for prediction is accurate and well prepared. Additionally, the project aims to implement advanced forecasting models, such as ARIMA, to effectively capture patterns like trends and fluctuations in crop prices. The project also focuses on presenting the prediction results in a clear, user-friendly, and visually understandable manner. By using graphical visualizations and structured outputs, the system ensures that users can easily interpret both historical trends and future predictions without requiring technical expertise. Ultimately, AgroSense aims to support better decision-making by providing timely and data-driven insights, helping users plan their selling strategies more effectively, improve income opportunities, and reduce uncertainty in agricultural market participation.

1.5 Scope Of Project

The scope of the AgroSense project extends beyond basic crop price prediction to various real world applications in the agricultural and economic domains. The system can be used as a decision support tool for farmers, traders, and agricultural stakeholders to understand market trends and plan their activities more effectively. By analyzing historical data and forecasting future prices, it can assist in better crop planning, storage decisions, and timing of market entry, thereby contributing to improved supply chain management. In addition, the project has applications in agricultural market analysis and policy planning. Government agencies and agricultural organizations can use such systems to monitor price trends, identify patterns, and make informed decisions related to pricing policies, subsidies, and market regulations. It can also be extended to support agri business companies, wholesalers, and retailers in demand forecasting, inventory management, and strategic planning, helping them align their operations with expected market conditions. Furthermore, the system can be adapted and scaled to include multiple crops, regions, and real time data integration, making it suitable for large scale agricultural analytics platforms. It can also be incorporated into mobile or web based applications to improve accessibility and usability for a wider audience. With further enhancements, AgroSense can be integrated with advanced technologies such as IoT and real time data streams, expanding its usability in smart agriculture, precision farming, and digital farming ecosystems.

CHAPTER 2

LITERATURE REVIEW

2.1 Existing Tools And Research

Forecasting Agricultural Commodity Prices Using Model Selection Framework

Agricultural commodity prices are highly volatile and influenced by various time dependent factors, making accurate prediction a challenging task. This study proposes a model selection framework that improves forecasting accuracy by combining multiple machine learning models. The approach focuses on extracting relevant time series features and selecting the most suitable model for prediction based on performance. By comparing different forecasting techniques and optimizing feature selection, the framework enhances reliability and reduces prediction error, making it useful for intelligent agricultural price prediction systems.

The methodology begins with collecting agricultural price data and converting it into a time series format. Feature extraction is performed to identify patterns such as trend, seasonality, and stationarity. A feature reduction step is then applied to eliminate redundant data and improve model efficiency. Multiple forecasting models such as ANN, SVR, and ELM are trained using the processed data. A model selection mechanism is used to choose the best-performing model based on accuracy for different forecast horizons. The final predictions are evaluated using error metrics like MAPE to ensure effectiveness.

Price Risk Early Warning Model for Agricultural Products Using Spatio Temporal Learning

Predicting agricultural price fluctuations requires understanding both temporal trends and spatial variations. This paper presents a spatio temporal deep learning model that captures complex relationships between price data, regions, and influencing factors. The proposed system integrates risk aware modeling to not only forecast prices but also assess uncertainty. By combining advanced learning techniques with multi dimensional data, the model provides accurate predictions and early warnings, contributing to better planning and decision making in agricultural systems.

The methodology involves collecting time series price data along with spatial and contextual information such as location based variations. The data is preprocessed and structured to capture both temporal and spatial dependencies. A deep learning model is then applied to learn patterns across time and regions simultaneously. The system incorporates a risk aware mechanism to evaluate uncertainty in predictions. Continuous model updating is performed using new data inputs, and the final output includes both predicted prices and associated risk levels for better analysis.

Table 2.2.1 Comparative Analysis and Research Gap

S.No	Existing Work	Methodology	Research Gap
1	AGMARKNET mandi portal	Displays daily mandi crop prices	No predictive analytics
2	eNAM digital market	Online trading platform for farmers and traders	Does not provide price forecasting or predictive analytics
3	Online grocery platforms	Show retail fruit and vegetable prices	Farmers lack market insights
4	Statistical price analysis	Uses historical data and statistical models	Limited ability to capture price trends
5	Agro Price Oracle (Proposed one)	Web scraping + ML prediction + dashboard	Provides price forecasting for farmers

2.2 Research Gap

Despite achieving good prediction accuracy, the model selection framework primarily focuses on selecting the best forecasting model from a predefined set of machine learning techniques. It requires training multiple models such as ANN, SVR, and ELM, which increases computational complexity and time. Additionally, the system is more model centric rather than user centric, as it emphasizes prediction performance but does not provide a simple, interactive interface for end users to easily interpret results. It also depends heavily on feature engineering and selection processes, which may limit scalability when applied to new datasets or different crops. Similarly, the spatio temporal deep learning model introduces advanced techniques for capturing complex relationships between agricultural factors, but it becomes highly complex and resource intensive. The architecture requires large amounts of structured data, including environmental and regional information, which may not always be readily available. Its complexity also makes it less practical for real time or small scale applications, and the interpretability of results can be challenging for non technical users. Moreover, the focus is more on risk modeling and large scale optimization rather than providing a lightweight, accessible forecasting system for everyday decision making. Therefore, there is a clear research gap in developing a system that is both accurate and practical one that provides reliable crop price predictions while being simple, efficient, and user friendly. Addressing these limitations, AgroSense aims to offer a streamlined solution using effective time series forecasting techniques with reduced complexity and an intuitive interface, making it more accessible and useful for real world applications.

CHAPTER 3

SYSTEM DESIGN

3.1 System Architecture

The system begins with the Data Sources, which act as the foundation of the entire architecture by providing raw agricultural price data. These sources may include online marketplaces, historical datasets, and publicly available pricing information. This raw data is then collected through the Web Scraping Layer, where automated scripts extract relevant price details at regular intervals. This layer ensures continuous and up-to-date data collection without manual intervention, enabling the system to work with real-time or near real-time information. Once the data is collected, it is stored in the Data Storage (Database), which serves as a centralized repository for all historical and current data. The database ensures data integrity, organization, and efficient retrieval for further processing. After storage, the data moves to the Data Processing & Feature Engineering stage, where it is cleaned, structured, and transformed. This step includes handling missing values, removing inconsistencies, and extracting meaningful features such as trends, seasonality, and patterns that are essential for accurate predictions. The processed data is then used in the Machine Learning Model Training phase, where forecasting models such as ARIMA are applied to learn patterns from historical data and generate future price predictions. To ensure efficient deployment and monitoring, the system incorporates an MLOps Pipeline (MLflow + Automation), which manages model training, versioning, evaluation, and automated updates. This pipeline ensures that the model remains accurate and adaptable over time as new data becomes available. Finally, the trained model is exposed through a REST API, which acts as a bridge between the backend and frontend, allowing external applications to request predictions. The results are displayed in the Web Dashboard, where users can visualize insights such as the buyer vs seller price gap. This interactive interface presents data in a clear and understandable manner, enabling users to analyze trends and make informed decisions effectively.

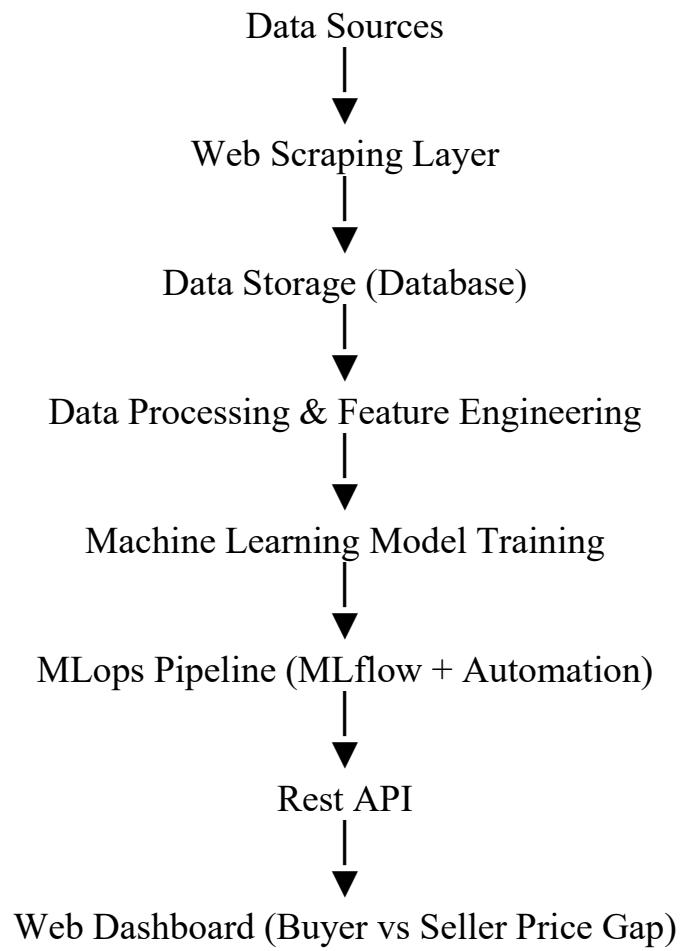


Figure 3.3.1 System architecture

3.2 Process Flow

The process begins when the system initiates data collection from online grocery platforms such as Zepto, BigBasket, and Instamart. The scraping module connects to these platforms and extracts the latest crop price information at regular intervals. This raw data is then immediately transferred into the system and stored in a database, where it builds a continuous historical record of price changes over time. Once stored, the system processes the collected data by cleaning inconsistencies, removing duplicates, and organizing it into a structured time based format. After preprocessing, the refined dataset is passed to the model stage, where the system either trains the forecasting model or uses the existing trained model to generate future price predictions. This entire operation is handled through an automated pipeline, ensuring that new data updates the model without manual effort. After predictions are generated, they are made accessible through a backend service. Whenever the frontend requests data, the system retrieves the latest predicted values and sends them to the dashboard. Finally, the results are displayed in a visual format where users can clearly observe trends and identify the gap between buyer and seller prices, enabling better understanding of market behavior and supporting informed decision making.

CHAPTER 4

TECHNOLOGY STACK

4.1 Frontend

The frontend of the AgroSense crop price prediction system is designed using Python based visualization techniques that present forecasting results in a clear and understandable format. Instead of using a web based interface, the system provides an interactive graphical output through plots generated using Matplotlib, which serves as the primary visualization component of the application. These graphical representations help users observe historical price trends and forecasted values effectively. In this project, the frontend accepts processed data from the backend forecasting module and displays the results in the form of time series graphs showing tomato price variations over a selected period. The visualization compares historical price data with predicted values generated by the ARIMA model, enabling users to easily analyze market behavior and expected future trends. The plotted graphs include properly labeled axes, legends, grid structures, and titles that improve readability and interpretation. The frontend component also displays summary outputs such as predicted price values for upcoming days and model performance information through the execution console. This approach ensures that users can understand both numerical predictions and graphical insights without requiring a complex web interface. By presenting the results in a structured and visual manner, the frontend improves the usability of the AgroSense system and supports better decision making based on forecasted agricultural price trends.

4.2 Backend

The backend of the AgroSense crop price prediction system is built using Python and serves as the core engine that drives data handling, forecasting, and system logic. It is responsible for managing the complete flow starting from collected raw data to generating final predictions. Data obtained through the scraping layer is first handled in the backend, where libraries like Pandas and NumPy are used extensively for cleaning, filtering, and structuring the dataset into a consistent time series format. This includes handling missing values, removing duplicates, and aligning data based on timestamps to ensure continuity in price records. Once the data is prepared, the backend performs feature-level processing such as resampling the data into daily or weekly intervals and identifying patterns required for forecasting. The ARIMA model is then applied as the core prediction technique, where the backend manages parameter selection, model fitting, and validation using historical price data. It ensures that the model captures trends and variations effectively to produce reliable future price predictions. The backend also supports retraining or updating the model whenever new data is added, maintaining the accuracy of predictions over time. In addition to model execution, the backend is designed to expose prediction results through a REST API, allowing seamless communication with the frontend dashboard. It handles request response cycles, where user queries trigger the retrieval of the latest predictions. The backend also manages intermediate storage of processed data and prediction outputs, ensuring quick access and efficient performance. Overall, it acts as the central processing unit of AgroSense, integrating data processing, machine learning, and system communication into a unified workflow.

4.3 AI Layer

The AI layer of the AgroSense crop price prediction system is responsible for analyzing historical agricultural price data and generating future price forecasts using statistical time series modeling techniques. In this project, the AutoRegressive Integrated Moving Average (ARIMA) model is used as the core prediction algorithm because it is highly effective for handling sequential data that varies over time. The model studies past tomato price trends collected from different platforms and identifies hidden patterns such as seasonality, fluctuations, and dependencies between previous observations and future values. Before applying the ARIMA model, the collected data is preprocessed using Python libraries such as Pandas and NumPy to ensure consistency and continuity in the time-series dataset. The cleaned dataset is then resampled into daily average price values to maintain uniform intervals required for forecasting. After preprocessing, the ARIMA model is trained on historical price data and used to predict future price values for the next thirty days. These predicted values help in understanding expected market behavior and assist users in making better decisions regarding crop selling strategies. Thus, the AI layer plays an important role in transforming raw agricultural price data into meaningful predictive insights by applying time-series forecasting techniques and statistical learning methods.

4.4 Visualization Engine

The visualization engine of the AgroSense crop price prediction system is responsible for presenting historical and forecasted price data in a graphical and easy to understand format. In this project, the visualization component is implemented using the Matplotlib library in Python, which helps generate clear and informative time series plots of tomato price variations. These visual graphs allow users to observe past price trends along with predicted future values produced by the ARIMA forecasting model. The visualization engine plots both historical daily average prices and forecasted prices on the same graph, enabling users to compare previous market behavior with expected future trends. Important graphical elements such as axis labels, titles, legends, markers, and grid lines are included to improve clarity and readability of the results. This graphical representation makes it easier for users to interpret prediction outcomes without requiring advanced technical knowledge. Thus, the visualization engine plays a significant role in transforming numerical forecasting results into meaningful visual insights, helping users better understand crop price movements and supporting informed decision making.

CHAPTER 5

IMPLEMENTATION

5.1 Problem Description Mode

During implementation, several practical challenges arise while building the AgroSense system. The primary difficulty lies in handling inconsistent and unstructured data collected from multiple online sources, where price formats, units, and update frequencies vary significantly. This requires careful preprocessing to ensure uniformity before analysis. Another challenge is maintaining continuity in time series data, as missing values or irregular intervals can affect the accuracy of forecasting models like ARIMA. Selecting appropriate model parameters is also non trivial, since improper tuning can lead to poor predictions. Additionally, ensuring that the system remains efficient while processing continuously updating data introduces complexity in managing storage, computation time, and scalability. These issues collectively highlight the need for a robust backend pipeline that can handle data variability, maintain data quality, and support reliable forecasting.

5.2 Execution Mode

In execution, the AgroSense system operates as an automated pipeline where each component performs a specific task in sequence. Data is periodically collected and immediately processed to maintain an updated dataset. The system then transforms this data into a structured time-series format and feeds it into the trained ARIMA model for generating predictions. The execution flow ensures that whenever new data is available, the model is either updated or used to produce fresh forecasts without manual intervention. The backend manages these operations efficiently and exposes the results through an API, allowing the frontend dashboard to fetch and display the latest predictions. This execution mode focuses on continuous operation, timely updates, and smooth integration between data processing, model prediction, and user interface, ensuring that the system delivers consistent and real time insights.

5.3 Working Model

The working model of AgroSense is built using real time and historical crop price data collected through web scraping from online grocery platforms such as Zepto, BigBasket, and Instamart. The scraped dataset consists of price values recorded over time, which forms the basis for time series analysis. In addition to scraped data, the system may also incorporate structured historical datasets to improve model stability and learning. Before applying any algorithm, the data undergoes preprocessing where missing values are handled using techniques such as replacing them with mean values or zero values depending on the context, and duplicate or inconsistent entries are removed to maintain data quality. For prediction, the system uses the ARIMA (AutoRegressive Integrated Moving Average) algorithm, which is well suited for time series forecasting. The model works by identifying patterns such as trends and seasonality in historical price data. Key parameters of ARIMA, including p (autoregressive term), d (differencing order), and q (moving average term), are carefully selected to achieve optimal performance. The system also considers error metrics such as Mean

Absolute Error (MAE) or similar statistical measures to evaluate prediction accuracy and fine tune the model. Data resampling techniques, such as converting irregular data into daily or weekly intervals, are also applied to maintain consistency in the dataset. By combining cleaned datasets, properly handled missing values, optimized model parameters, and continuous evaluation through error metrics, the working model is able to generate reliable future price predictions. This structured approach ensures that the system maintains accuracy, stability, and adaptability, making it effective for real world agricultural price forecasting.

Scraping

The screenshot displays a VS Code interface with a file explorer on the left, a code editor in the center, and a terminal at the bottom. The file explorer shows a project named 'AGROSENSE' with files like '.venv', 'arima.py', 'bigbasket.py', 'instamart_tomatoes.csv', 'instamart.py', 'sample_instamart.py', and 'tomato_dataset.csv'. The code editor shows the 'bigbasket.py' script, which uses Selenium to scrape data from the BigBasket website. The terminal shows the command to run the script and the successful execution, resulting in the 'Clean tomato dataset updated!' message.

```

1  from selenium import webdriver
2  from selenium.webdriver.chrome.service import Service
3  from webdriver_manager.chrome import ChromeDriverManager
4  from selenium.webdriver.common.by import By
5  import time
6  import csv
7  from datetime import datetime
8  import re
9  import os
10
11 # Setup driver
12 options = webdriver.ChromeOptions()
13 options.add_argument("--start-maximized")
14
15 driver = webdriver.Chrome(
16     service=Service(ChromeDriverManager().install()),
17     options=options
18 )
19
20 # Open BigBasket tomato page
21 driver.get("https://www.bigbasket.com/ps/?q=tomato")
22 time.sleep(5)
23
24 # Scroll to load products
25 for _ in range(3):
26     driver.execute_script("window.scrollTo(0, 1000);")

```

```

PS C:\Users\Admin\OneDrive\Pictures\Desktop\AgroSense> & 'c:\Users\Admin\OneDrive\Pictures\Desktop\AgroSense\.venv\Scripts\python.exe' 'c:\Users\Admin\OneDrive\Pictures\Desktop\AgroSense\bigbasket.py'
KeyboardInterrupt
PS C:\Users\Admin\OneDrive\Pictures\Desktop\AgroSense> c:: cd 'c:\Users\Admin\OneDrive\Pictures\Desktop\AgroSense'
Clean tomato dataset updated!
PS C:\Users\Admin\OneDrive\Pictures\Desktop\AgroSense>

```

Figure 5.3.1 Scraping initiated and final output is csv

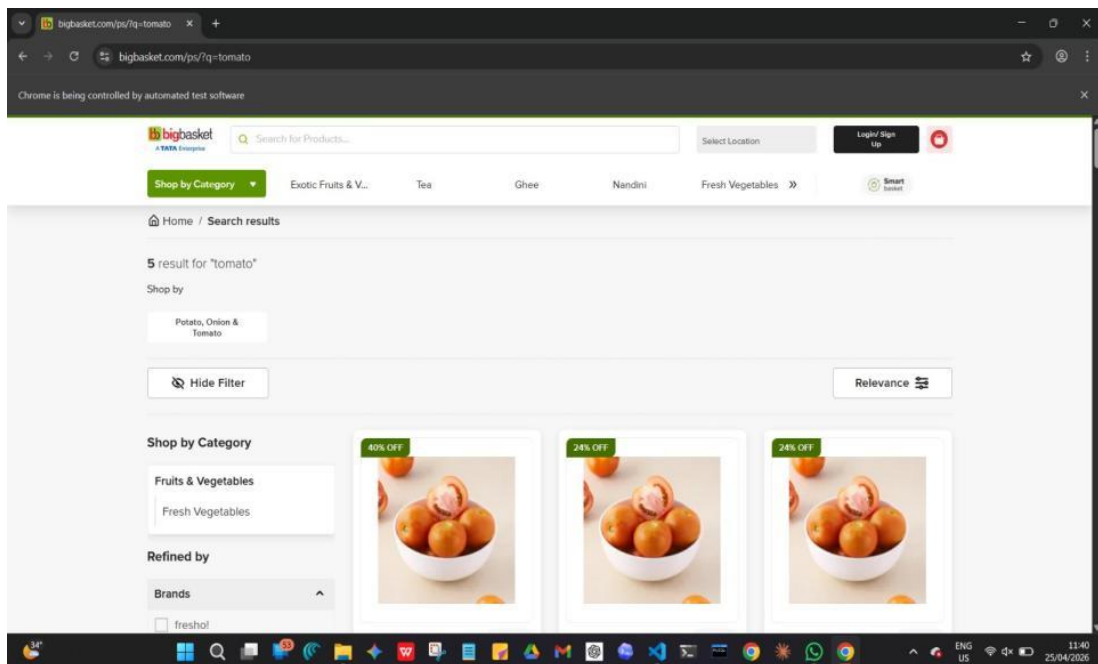


Figure 5.3.2 Automated test software scrolling through products

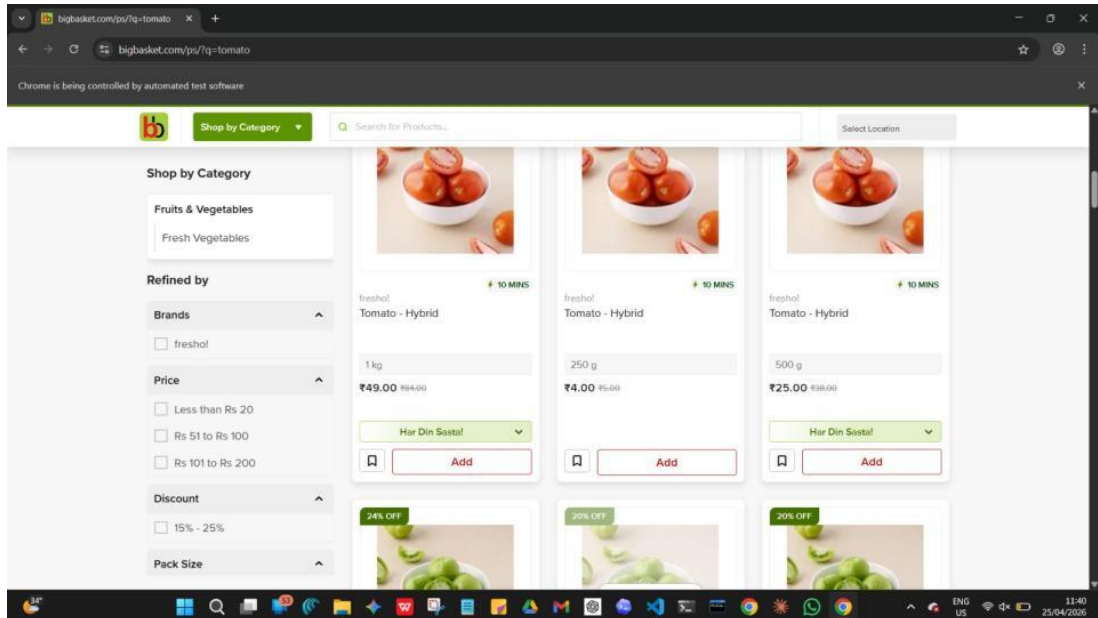


Figure 5.3.3 Automated test software getting through the end of the required crop

CHAPTER 6

RESULT

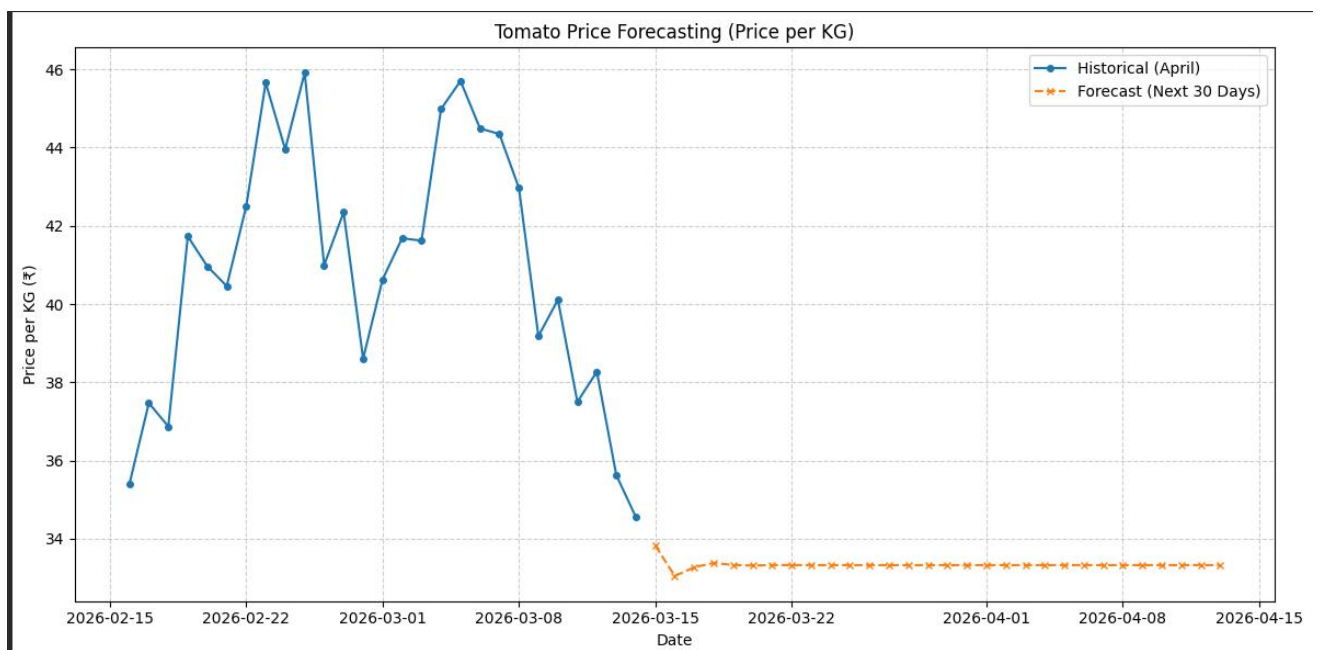


Figure 6.1 : Showing rise and fall of prices from 2026-02-16 to 2026-03-15

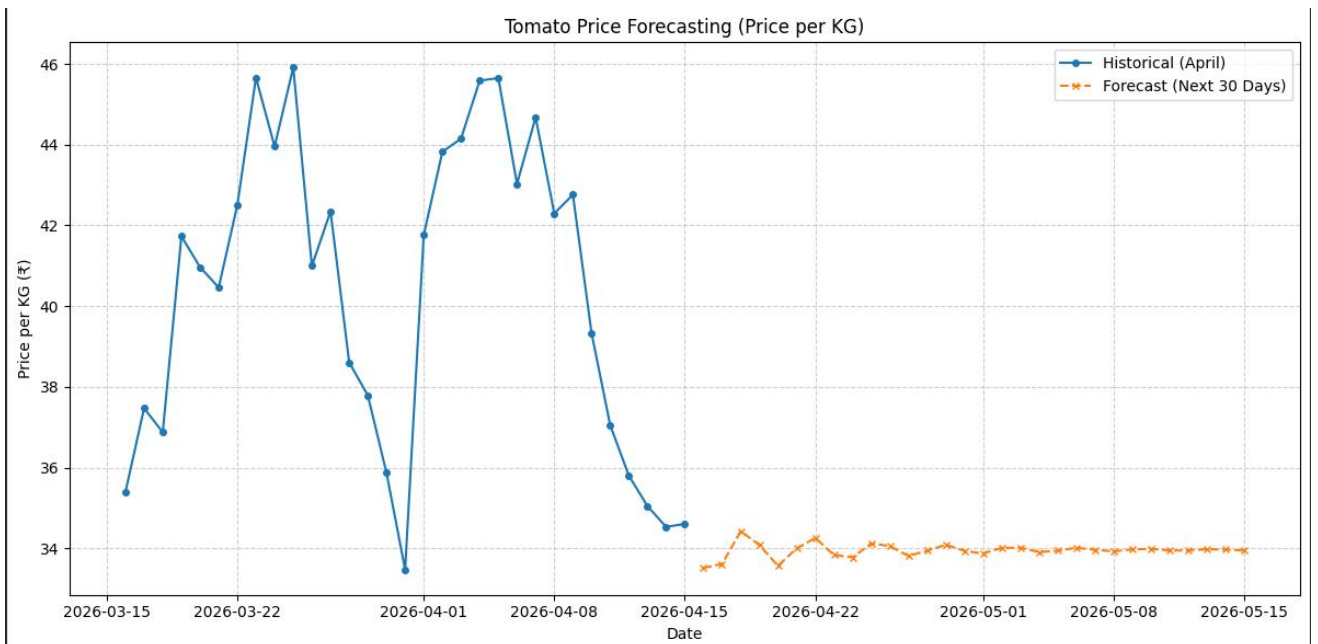


Figure 6.2: Showing rise and fall of prices from 2026-03-16 to 2026-04-15

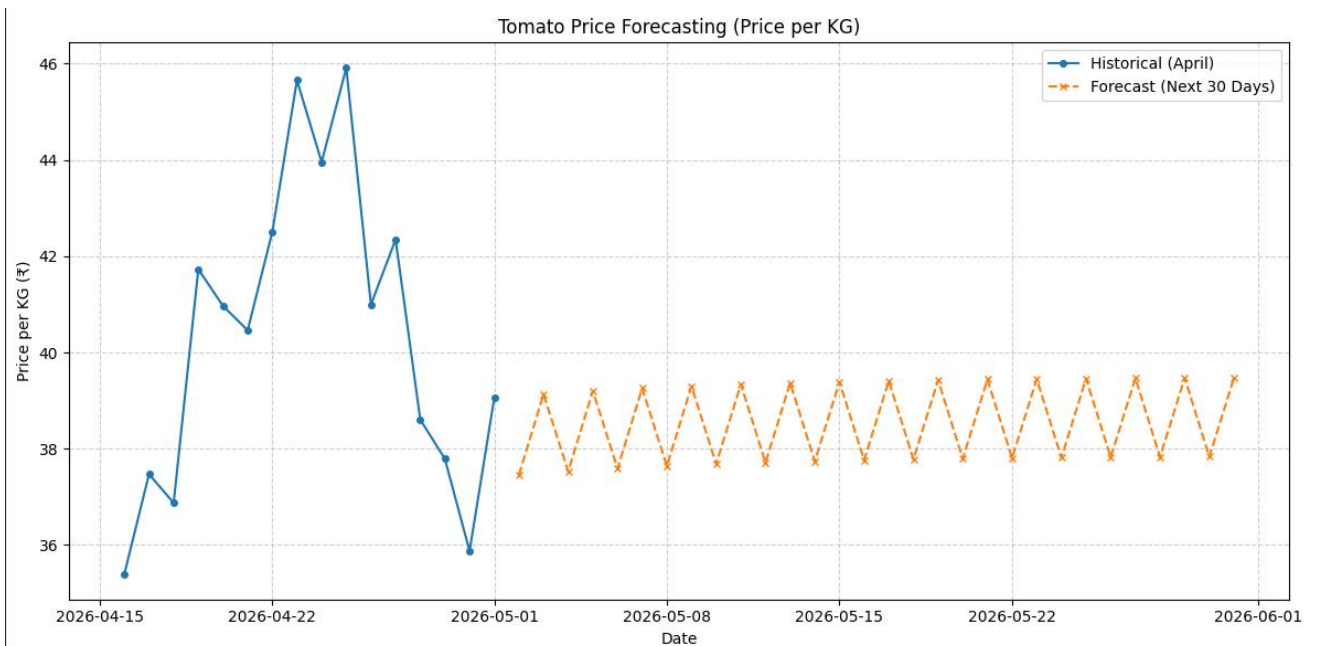


Figure 6.3: Showing rise and fall of prices from 2026-04-16 to 2026-05-01

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Conclusion

The AgroSense project presents a comprehensive approach to crop price prediction by integrating data collection, preprocessing, machine learning, and visualization into a unified system. The project effectively demonstrates how raw agricultural price data, when properly processed and analyzed, can be transformed into meaningful insights that support better decision making. By utilizing Python based backend processing along with libraries such as Pandas and NumPy, the system ensures efficient handling of large and unstructured datasets. The implementation of the ARIMA model plays a crucial role in capturing time series patterns such as trends and fluctuations, enabling the generation of reliable future price predictions. A key strength of the system lies in its structured pipeline, where each stage from data scraping to final visualization works in coordination to deliver accurate results. The preprocessing techniques applied, including handling missing values using mean or zero replacement and maintaining consistent time intervals through resampling, significantly improve the quality of input data. This, in turn, enhances the performance of the forecasting model. The integration of a REST API and dashboard further ensures that the predicted results are accessible and easy to interpret, allowing users to understand price trends and variations effectively. Overall, AgroSense successfully bridges the gap between raw data and actionable insights by combining data engineering and machine learning. It highlights the importance of predictive analytics in agricultural systems and demonstrates how technology can be used to improve efficiency, transparency, and understanding of market behavior. The project serves as a strong foundation for developing more advanced and scalable solutions in the domain of agricultural forecasting.

7.2 Future Work

Although AgroSense provides an effective crop price prediction system, it can be enhanced by supporting multiple crops and regions to improve scalability. Incorporating real-time data streams and advanced models like LSTM can further increase prediction accuracy. Including external factors such as weather, demand, and transportation costs would make predictions more realistic. The system can also be improved with better automation, interactive dashboards, and a mobile-friendly interface. In the future, it can evolve into a smart agriculture platform by integrating IoT and enabling end-to-end decision support.

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